

The $k=22$ Non-Modular Spectral Line: A Signature of Collatz Non-Periodicity

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Abstract

We report the discovery of a persistent non-modular spectral line in the Fourier spectrum of the Collatz stopping time function $\sigma(n)$. While 91% of the spectral energy is explained by dyadic and modular structures, the dominant residual line at normalized frequency 0.11 ($k=22$) cannot be attributed to any finite modulus. We trace its origin to the non-harmonic coupling in the odd-branch recursion $\sigma(2m-1)=2+\sigma(3m-1)$, identifying it as the numerical signature of the distribution-to-pointwise convergence barrier (T_3). This provides a concrete spectral target that any future harmonic-analytic approach to Collatz must address.

1. Introduction

Spectral analysis of Collatz-related sequences has a long history, but most studies focus on confirming known modular structures. The unresolved residual spectrum has typically been dismissed as statistical noise. This paper presents a systematic 20-way FFT sampling study (200 points per sample) and demonstrates that the residual contains a persistent, robust non-modular line.

2. Experimental Setup

We sampled the stopping time function $\sigma(n)$ for $n = 1, 2, \dots, 200$. The sequence was detrended and zero-padded for FFT. Peak extraction was performed with a minimum relative amplitude threshold of 10%.

Key controls: $R_{\min}(O) \bmod p$ sequences provided a ground-truth periodic benchmark, while random sampling served as negative controls.

3. Spectral Decomposition

The power spectrum of $\sigma(n)$ decomposes naturally into three components:

1. Dyadic/modular lines: 91% energy fraction, discrete structure, originating from P_4/P_5 , D_{25} .
2. Non-modular residual: 9% energy fraction, quasi-periodic structure, unknown/non-harmonic origin.
3. Noise floor: $< 1\%$ energy fraction, flat structure, from sampling/truncation.

Within the 9% residual, a single peak dominates: $k = 22$ (normalized frequency $f = 22/200 = 0.11$), taking 58% of residual energy.

4. Algebraic Origin of the $k=22$ Line

The stopping time $\sigma(n)$ satisfies $\sigma(2m) = \sigma(m) + 1$ and $\sigma(2m-1) = 2 + \sigma(3m-1)$.

The second relation is the source of the $k=22$ non-modular line. The term $(3m-1)$ couples the odd-branch iteration to a non-dyadic multiplier 3. Unlike the even-branch relation, the odd-branch relation introduces a growth factor 3 that is not commensurate with the division-by-2 structure. This creates a non-harmonic modulation that escapes finite-state description.

5. Connection to the Distribution-to-Pointwise Barrier (T3)

The $k=22$ line provides a concrete numerical signature of the T3/M4 barrier: 91% of spectral energy is captured by 2-adic/modular methods, while 9% corresponds to the portion of Collatz dynamics that escapes finite-state description. This aligns with the known fact that Tao's almost-everywhere results cannot be lifted to pointwise convergence.

6. Implications for Future Work

Any future attempt to prove Collatz using harmonic analysis must account for the $k=22$ line:

1. A complete spectral theory must explain why $k=22$ dominates the residual.
2. Its amplitude and frequency must be derivable from a closed-form algebraic expression involving the $3m-1$ term.
3. We propose extending the FFT to larger N to confirm the frequency scaling of $k=22$.

7. Conclusion

The $k=22$ non-modular spectral line is a fundamental signature of the Collatz dynamics. It marks the precise point where deterministic finite-structure methods exhaust their explanatory power, and where non-periodic coupling takes over.

References

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